

JEE MAIN REVISION MASTER: SOLUTIONS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 188 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Colligative Properties & Van 't Hoff Factor	103 PYQs	Colligative
Very High	Raoult's Law & Vapor Pressure (Ideal & Non-ideal)	88 PYQs	Vapor Pressure

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Colligative Properties & Van 't Hoff Factor	$\frac{P^\circ - P}{P^\circ} = i \cdot x_B, \quad \Delta T_b = i K_b m, \quad \Delta T_f = i K_f m, \quad \Pi = i C R T, \quad i = 1 + (n - 1)\alpha$
Raoult's Law & Vapor Pressure (Ideal & Non-ideal)	$P_{\text{total}} = P_A^\circ x_A + P_B^\circ x_B, \quad y_A = \frac{P_A}{P_{\text{total}}} = \frac{P_A^\circ x_A}{P_{\text{total}}}, \quad P = K_H \cdot x \text{ (Henry)}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Colligative Properties & Van 't Hoff Factor (Very High Yield — 103 PYQs)

Core Principle: Physical properties depending solely on solute particle count, cryoscopic/ebullioscopic constants, and Van 't Hoff factor i .

■ **Caution / JEE Pitfall:** When determining relative boiling/freezing points of salt solutions, compare effective molality $i \times m$, NOT just nominal molality m !

■ **Speed Trick:** For dimerization/association of n molecules: $i = 1 - \left(1 - \frac{1}{n}\right)\alpha$ implies $i < 1$ (e.g. benzoic acid in benzene, $i \approx 0.5$).

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "At 27°C , 0.1M , 1L $\text{K}_2\text{Cr}_2\text{O}_7$ aqueous solution and 0.1M , 1L FeCl_3 aqueous solution are placed in a container separated by a s..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "W' g of a non-volatile electrolyte solid solute of molar mass 'M' gmol^{-1} when dissolved in 100 mL water, decreases vapour pressure..." [Ans: D]
- [2026 Evening Shift] Q#1 (Numerical): "The osmotic pressure of a living cell is 12 atm at 300 K . The strength of sodium chloride solution that is isotonic with the living cell at ..." [Ans: 15]

3.2 Raoult's Law & Vapor Pressure (Ideal & Non-ideal) (Very High Yield — 88 PYQs)

Core Principle: Partial pressures of volatile mixtures, Dalton's law in vapor phase (y_A), and thermodynamic criteria for deviations ($\Delta H_{\text{(mix)}}$, $\Delta V_{\text{(mix)}}$).

■ **Caution / JEE Pitfall:** Mixture of Chloroform and Acetone exhibits **NEGATIVE** deviation from Raoult's law ($\Delta H_{\text{(mix)}} < 0$, $\Delta V_{\text{(mix)}} < 0$) due to intermolecular hydrogen bonding.

■ **Speed Trick:** Higher Henry's law constant K_H at a given pressure signifies **LOWER** solubility of the gas in liquid.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "Given below are two statements : Given : Molar mass of C,H,O,Cl are 12,1,16 and 35.5gmol⁻¹ , respectively Statement I : In 30%(w/w)..." **[Ans: N/A]**
- **[2026 Morning Shift] Q#1 (MCQ):** "At T(K), 2 moles of liquid A and 3 moles of liquid B are mixed. The vapour pressure of ideal solution formed is 320 mm Hg . At this stage, ..." **[Ans: C]**
- **[2026 Morning Shift] Q#1 (MCQ):** "Given below are two statements : Statement I : The Henry's law constant K_H is constant with respect to variations in solution's concent..." **[Ans: B]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 28th January Morning Shift (Q#1)

At T(K), 2 moles of liquid A and 3 moles of liquid B are mixed. The vapour pressure of ideal solution formed is 320 mm Hg . At this stage, one mole of A and one mole of B are added to the solution. The vapour pressure is now measured as 328.6 mm Hg . The vapour pressure (in mm Hg) of A and B are respectively:

- (A) 400, 300
- (B) 600, 400
- (C) 500, 200 **[Correct]**
- (D) 300, 200

Correct Answer: C

Solution Analysis: For an ideal solution, by Raoult's law: $P_{\text{(total)}} = x_A P_A^0 + x_B P_B^0$ where P_A^0, P_B^0 are vapour pressures of pure liquids A and B. Step 1: First mixture (2 mol A + 3 mol B) Total moles = 2+3=5 $x_A = (2 / 5), x_B = (3 / 5)$ Given $P_{\text{(total)}} = 320$ mm Hg: $320 = (2 / 5)P_A^0 + (3 / 5)P_B^0$ Multiply by 5: $1600 = 2P_A^0 + 3P_B^0$...(1) Step 2: After adding 1 mol A and 1 mol B New moles: A =3 , B =4 , total =7 $x_A = (3 / 7), x_B = (4 / 7)$ Given new vapour pressure =328.6 mm Hg: $328.6 = \frac{3}{7}P_A^0 + \frac{4}{7}P_B^0$

Benchmark Problem 2: JEE Main 2026 (Online) 24th January Morning Shift (Q#1)

'W' g of a non-volatile electrolyte solid solute of molar mass 'M' g mol^{-1} when dissolved in 100 mL water, decreases vapour pressure of water from 640 mm Hg to 600 mm Hg. If aqueous solution of the electrolyte boils at 375 K and K_{b} for water is $0.52 \text{ K kg mol}^{-1}$, then the mole fraction of the electrolyte solute (x_2) in the solution can be expressed as (Given : density of water = 1 g/mL and boiling point of water = 373 K)

- (A) $(16 / 2 \cdot 6) \times (W / M)$
- (B) $(1 \cdot 3 / 8) \times (M / W)$
- (C) $(2 \cdot 6 / 16) \times (M / W)$
- (D) $(1 \cdot 3 / 8) \times (W / M)$ **[Correct]**

Correct Answer: D

Solution Analysis: Boiling point elevation: $\Delta T_{\text{b}} = 375 - 373 = 2 \text{ K}$ Mass of water = $100 \text{ mL} \times 1 \text{ g mL}^{-1} = 100 \text{ g} = 0.1 \text{ kg}$ Molality: $m = (\text{moles of solute} / \text{kg of solvent}) = (W/M) / 0.1 = 10(W/M)$ For electrolyte (NCERT): $\Delta T_{\text{b}} = iK_{\text{b}}m$ $2 = i(0.52)(10(W/M)) = i(5.2(W/M))$ $i = (2 / 5.2) \cdot (M/W) = (5/13) \cdot (M/W)$ Vapour pressure lowering: $\frac{\Delta p}{p^\circ} = (640 - 600 / 640) = (40 / 640) = (1 / 16)$ For electrolyte (NCERT): $\frac{\Delta p}{p^\circ} = ix_2$ So, $x_2 = \frac{1/16}{i} = \frac{1}{16} \cdot \frac{1}{i}$